PD1 · STATIC-PDELTA

# Tension stiffening via P-Delta analysis (CSI 1-016)

Production P-Delta static path (fixed gravity reference) — tension-stiffened deflection & moment vs Timoshenko closed form

**PASS**

error −0.0505% · tol 0.1%

## §2 TECHNICAL FEATURES TESTED

- **STRIX engine 2-node frame element** + geometric stiffness **$K_G$** (P-Delta)
- Gravity reference: axial-tension load case fixes $K_G$ from its member axial forces
- Target load case: transverse load solved on **$K + K_G$**
- Same code path as the production single-load-case P-Delta analysis (`generatePyForPDeltaLoadCase`)
- Deflection & moment vs **exact tension-stiffened beam** (Timoshenko)

## §3 PROBLEM DESCRIPTION

A slender pin-roller tie-rod carries a small transverse uniform load together with a large axial tension pulling the roller end. Tension measurably **stiffens** the beam against the transverse load — the deflection and moment both shrink relative to the no-tension case — the classic P-Delta tension-stiffening problem with a closed-form solution (Timoshenko 1956). STRIX solves this with its production **P-Delta static analysis with a fixed gravity reference**: the axial-tension load case is solved first and its member axial forces fix the geometric stiffness $K_G$; the transverse load case is then solved on $K + K_G$. This is the same path the app uses for P-Delta load cases on the STRIX engine, so the benchmark exercises the production code path directly. The reference problem and closed form are those of CSI SAP2000 verification example 1-016.

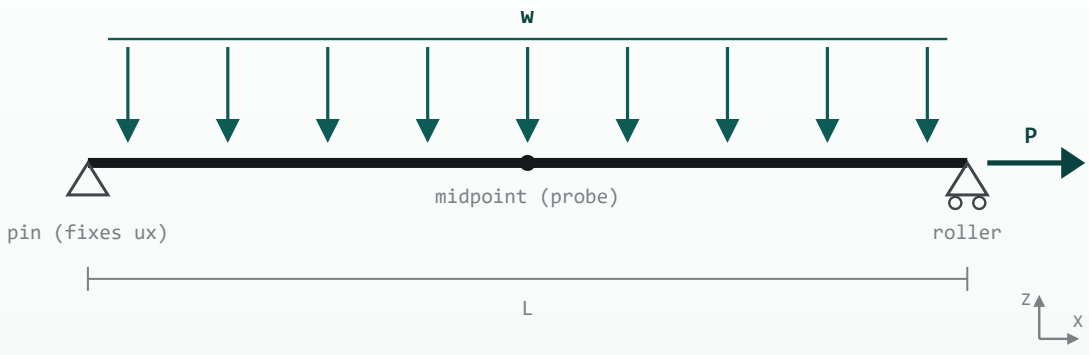



**Figure PD1.** Simply-supported tie-rod (pin–roller, moment-free both ends) under a transverse uniform load w plus an axial tension P pulling the roller end; midpoint deflection/moment are probed.

## §4 GEOMETRY, PROPERTIES & LOADING

| GEOMETRY | | MATERIAL & LOADING | | SUPPORTS & ANALYSIS | |
|---|---|---|---|---|---|
| Span L | 300 in | E | 30000 ksi | Supports | pin + roller |
| Section | 3 × 3 in (square) | Axial tension P | 20.25 kip | Gravity reference | axial-tension LC → K_G |
| Inertia I | 6.75 in⁴ | Transverse w | 0.002 kip/in | Target LC | transverse w on K + K_G |
| Tension param. u | 1.5 | Self-weight | off | Engine | STRIX engine (P-Delta, fixed gravity reference) |

## §5 REFERENCE SOLUTION

**Uz (no tension): -5*w*L^4/(384*E*I) = -1.041667 in**

**My (no tension): w*L^2/8 = 22.5 kip-in**

**Uz (w/ tension): Timoshenko eq.43: (5wL^4/384EI) * [1/cosh(u) - 1 + u^2/2] / [(5/24)u^4], u=sqrt(PL^2/4EI) = -0.543305 in**

**My (w/ tension): Timoshenko eq.45: (wL^2/8) * 2(cosh(u)-1)/(u^2*cosh(u)) = 11.498079 kip-in**

*CSI SAP2000 Software Verification Example 1-016 'Tension Stiffening Using P-Delta Analysis'; independent reference = Timoshenko, Strength of Materials Part II, 1956, eq. 23 p.28 and eqs. 43/45 p.43. Reproduced from scratch by pd1Theory() to u=1.5, deflWithIn=-0.5433047259, momentWithKin=11.4980793, matching CSI's published -0.54330 in / 11.498 k-in exactly (exact unit conversion of the same physical problem, dimensionless u is unit-invariant).*

## §6 RESULTS COMPARISON

| RESPONSE QUANTITY | PROBE | STRIX | REFERENCE | DIFF | VERDICT |
|---|---|---|---|---|---|
| Uz (no tension) (in) | N17.uz | -1.041928 | -1.041667 | -0.0251% | PASS |
| My (no tension) (kip-in) | E17.Miy | 22.500021 | 22.5 | +0.000094% | PASS |
| Uz (w/ tension) (in) | N17.uz | -0.543579 | -0.543305 | -0.0505% | PASS |
| My (w/ tension) (kip-in) | E17.Miy | 11.492544 | 11.498079 | -0.0481% | PASS |

## §7 P-DELTA MESH CONVERGENCE (MIDPOINT DEFLECTION)

| MESH | UZ (IN) | DIFF |
|---|---|---|
| 2 elem | -0.595308 | -9.5716% |
| 4 elem | -0.556573 | -2.4421% |
| 8 elem | -0.546682 | -0.6217% |
| 16 elem | -0.5442 | -0.1648% |
| 32 elem | -0.543579 | -0.0505% |

The no-tension baseline (32 elements, P-Delta off) matches the Euler-Bernoulli closed form to 0.025% in deflection and 0.0001% in moment; the production frame element carries the section shear area (5/6·A), and the shear-deformation term $48EI/(5GA_sL^2)$ of a simply-supported beam under uniform load is 0.025% for this tie-rod. The **tension-stiffened** case is genuinely mesh-sensitive: the discrete geometric stiffness recovers the hyperbolic tension-stiffened shape only as elements are added. The midpoint deflection error falls from −9.57% (2 elements) to −2.44%, −0.62%, −0.165% and −0.050% (32 elements) — roughly a factor of four per mesh doubling, i.e. close to quadratic convergence. At the finest mesh STRIX reproduces the exact Timoshenko tension-stiffened deflection and moment to `about 0.05%`, confirming both that the P-Delta pipeline converges to the right physics and that it reduces to the plain linear-static answer when the axial tension is absent.

## §8 CONCLUSION

**PASS** STRIX reproduces the reference solution to **−0.0505%** (tolerance 0.1%). The tested element and its formulation are verified against the reference.

## §9 REFERENCES & ANALYSIS FILES

1. Timoshenko, S. — *Strength of Materials, Part II.* Eq. 23 p.28 (tension parameter u) and eqs. 43/45 p.43 (tension-stiffened deflection and moment).
2. CSI — *SAP2000 Software Verification, Example 1-016:* Tension Stiffening Using P-Delta Analysis (independent reference, Timoshenko 1956).
3. STRIX headless harness: `generatePyForPDeltaLoadCase → STRIX engine (static-pdelta route, fixed gravity reference) → node_disp.out · elem_force.out`; baseline without P-Delta via `generatePyForLoadCase`.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · PD1/` `sha256 (pending publish)`